

Shamita Goyal

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Education

University of California, San Diego

B.S. Cognitive Science, Human-Computer Interaction and Design, GPA: 3.87

San Diego, CA

Graduated Sept 2026

- **Relevant Coursework:** Data Structures, Algorithms, Databases, Advanced Python

Technical Skills

Languages: Java, Python, SQL, JavaScript, TypeScript, HTML/CSS, R, SASS/SCSS, Tailwind CSS

Technologies & Frameworks: React, Next.js, Express, MongoDB, Node.js, REST APIs, WebSockets, PostgreSQL, Firebase, Git/GitHub, Docker, CI/CD, Figma, Vercel, Netlify, Electron, Ionic, Supabase

Experience

Research Assistant Intern

Sept 2026 – Present

Healthcare Robotics Lab, UC San Diego CSE

San Diego, CA

- Building a Next.js and MongoDB caregiver dashboard for CARMEN with an Ollama-powered feedback loop that personalizes robot responses based on patient interactions.

Software Developer Researcher

Jan 2026 – Aug 2026

- Shipped **4 UI features** for CARMEN, an assistive dementia-care robot, using React and TypeScript; tested features with simulated users.
- Led migration of the frontend from Angular/Ionic to React, Vite, and TypeScript, with Docker and WebSockets; reduced codebase size by **24%**.
- Built a mock-data layer and **10 unit tests** to enable backend-independent development and testing.

Software Developer Research Intern

Jun 2025 – Feb 2026

Foundation Interface Lab, UC San Diego Design Lab

San Diego, CA

- Extended Meridian, an open-source Next.js/Zustand framework for customizable data interfaces; rebuilt its toolbar features and used **Rollup** to package the framework as an **NPM package**.
- Built a Next.js/Firebase **demo app** ingesting 80 Busan restaurants via **3 Tripadvisor API endpoints**. Added a Gemini-powered AI chat that converts natural-language requests into dynamic Meridian UI specifications.

Lead Software Developer

Feb 2025 – Jun 2026

Design Co, UC San Diego

San Diego, CA

- Led a 12-person cross-functional team to build and launch production websites, including (**DF26**, **Stride25**, **Up-Grade25**) using Next.js, React, TypeScript, and SCSS for a 3,000+ member organization.
- Modernized legacy Next.js codebases with SSR, improving site load times by up to **60%** while boosting SEO, Lighthouse scores, accessibility, and Core Web Vitals; mentored 2 developers through codebase modernization.
- Built event and program platforms reaching **550+** active users, **2.1K** site views, and **100+** registrations; led technical workshops for 1,000+ members.

Frontend Developer & UI Design Lead

Oct 2024 – Dec 2025

Data Science Student Society, UC San Diego

San Diego, CA

- Led development of 3 sites (**TritonBall**, **DS3**, **DS3 Consulting**) in React, TypeScript, Tailwind CSS, and Figma, taking **15+** pages from prototype to production and lifting homepage engagement **75%**.
- Built a shared design system with 9 developers for nonprofit clients: **8** reusable components across **10+** pages.

Projects

Solar IQ | React, TypeScript, D3.js/ECharts, Three.js, Python, PostgreSQL

DataHacks 2026, 1st Place

- Engineered a full-stack solar analytics platform processing **37K+** permit records with NREL, EIA, and Census data to estimate ZIP-level solar savings and visualize nationwide trends; won 1st place in the Zen Power sponsor track among 109 teams.

Gist | React, TypeScript, Tailwind CSS, Vite, FastAPI, Python, Google Gemini API, Pydantic

- Built a full-stack research assistant that processes PDFs into interactive knowledge maps and supports document-grounded chat using FastAPI and Gemini API.